STAC WARS PROBLEM SET 04

ASTRA

Mining the Outer Rim

A robotic asteroid-water supply concept for a future space economy.

MINE IN SPACE. SUPPLY IN SPACE. | TEAM: IIT MONAL



The problem is not mining. It is the cost.

The Challenge

- ✕ **Earth Launch Penalty: High cost per kilogram.**
- ✕ **Harsh Environment: Low gravity operations.**
- ✕ **Delivery Math: Transport costs outstrip resource value.**

The ASTRA Proposal

- ✓ **Targeting: Water-bearing near-Earth asteroids.**
- ✓ **Extraction: Robotic, modular thermal systems.**
- ✓ **Distribution: Supply directly to orbital customers.**

CORE THESIS

Deliver a useful resource where it is needed—not back to Earth.

Robotic Water-Mining Architecture

Every stage must function reliably for the mission to remain viable.

01

Launch & Transfer

Reach selected near-Earth asteroid.

02

Anchor & Drill

Grip the surface; collect material.

03

Thermal Extraction

Heat material to release vapour.

04

Condense & Store

Capture vapour; filter water.

05

Deliver in Space

Transfer to customer depot.

Power: Solar + Storage

Control: Autonomy + Fault detection

Comms: Delayed Earth links

Resilience: Safe mode + backups

What is proven vs. What is proposed

DEMONSTRATED BUILDING BLOCKS

- **Sample Return: OSIRIS-REx returned Bennu sample (2023).**
- **Navigation: Robotic spacecraft operate near small bodies.**
- **ISRU: Agencies actively research in-situ resource processing.**

NOT YET DEMONSTRATED

- **Scale:** Profitable large-scale asteroid extraction.
- **Logistics:** Routine commercial delivery of water.
- **Economics:** Verified cost per kilogram competitive with alternatives.

Three Cost Drivers

The savings are hypotheses to test—not guaranteed results.

01 Launch & Transfer

COST PRESSURE

Mass, launch and travel time.

DESIGN RESPONSE

Lightweight hardware; efficient trajectory.

02 Mining & Processing

COST PRESSURE

Drilling, heating and repairs.

DESIGN RESPONSE

Modular tools; solar-powered heating.

03 Failure & Ops

COST PRESSURE

Breakdowns and mission delays.

DESIGN RESPONSE

Autonomous fault detection; backups.

Who pays, and why buy?

THE PRODUCT

Filtered, usable water for space operations.

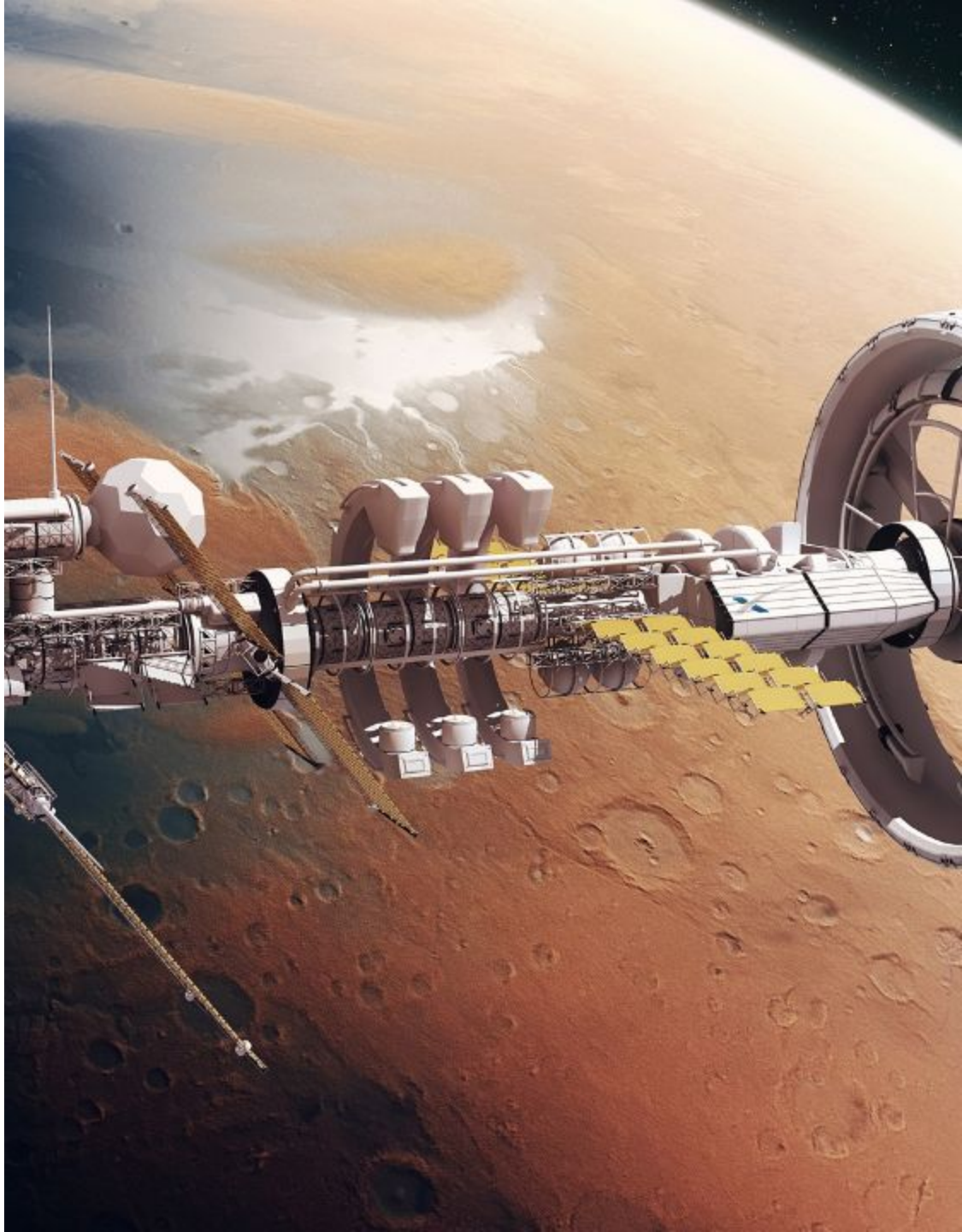
TARGET CUSTOMERS

Orbital depots and lunar logistics providers.

REVENUE MODEL

Pre-booked supply contracts; payment per kg.

> **Requirement: Customer price ≥ full delivered cost.**



Execution Roadmap

Manage risks via phased validation before scaling up.

01

Identify

Study a suitable asteroid target.

02

Test

Validate anchoring and extraction.

03

Demonstrate

Measure actual water yield and energy.

04

Scale

Proceed only if economics work.

ASTRA

MINE IN SPACE. SUPPLY IN SPACE.

Scale only when the economics work.

Image Sources



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